ON A REGULAR SCHEME OF DIMENSION AT MOST TWO A COHERENT MODULE IS REFLEXIVE EXACTLY WHEN IT IS FINITE LOCALLY FREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 2336-2345 and proof span 2347-2368 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof splits into the three possible values of the dimension of the local ring and refers to the cited statements by their hypotheses; the author's wording is preserved.

Editorial note. 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A finite locally free coherent module is reflexive. Used at proof lines 2347-2348. Gives the easy direction.
- A coherent module whose stalks are free is finite locally free, cited from the algebra chapter. Cited at proof lines 2350-2352. Reduces the converse to freeness of the stalk at each point.
- Local rings of dimension zero are fields and local rings of dimension one that are regular are discrete valuation rings, with torsion free finite modules over them free, cited from the algebra and more-algebra chapters. Cited at proof lines 2353-2359. Disposes of the points of dimension at most one.
- A reflexive module over a normal local ring has depth at least two, and a maximal Cohen-Macaulay module over a regular local ring is free, cited from the more-algebra and algebra chapters. Cited at proof lines 2360-2367. Disposes of the points of dimension two.

Source reference. Original lemma 00PD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 927–927. Same-chapter reference used in the retained author text at source line(s) 1819, 2358, 3021, 7023. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers

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Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Same-chapter reference used in the retained author text at source line(s) 1813, 2352. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00NT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 871–871. Same-chapter reference used in the retained author text at source line(s) 2367. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0AUW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13664–13664. Same-chapter reference used in the retained author text at source line(s) 1822, 2293, 2361. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0AY6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13779–13779. Same-chapter reference used in the retained author text at source line(s) 2275, 2364. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a regular scheme of dimension ≤ 2. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. The following are equivalent*

(1) *$\mathcal{F}$ is reflexive,*
(2) *$\mathcal{F}$ is finite locally free.*

Proof. It is clear that a finite locally free module is reflexive. For the converse, we will show that if $\mathcal{F}$ is reflexive, then $\mathcal{F}_x$ is a free $\mathcal{O}_{X,x}$-module for all $x \in X$. This is enough by Algebra, Lemma 00NX and the fact that $\mathcal{F}$ is coherent. If $\dim(\mathcal{O}_{X,x}) = 0$, then $\mathcal{O}_{X,x}$ is a field and the statement is clear. If $\dim(\mathcal{O}_{X,x}) = 1$, then $\mathcal{O}_{X,x}$ is a discrete valuation ring (Algebra, Lemma 00PD) and $\mathcal{F}_x$ is torsion free. Hence $\mathcal{F}_x$ is free by More on Algebra, Lemma 0AUW. If $\dim(\mathcal{O}_{X,x}) = 2$, then $\mathcal{O}_{X,x}$ is a regular local ring of dimension 2. By More on Algebra, Lemma 0AY6 we see that $\mathcal{F}_x$ has depth ≥ 2. Hence $\mathcal{F}_x$ is free by Algebra, Lemma 00NT. □